

ARUN SAGAR POTHULAPALLY

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PROFESSIONAL SUMMARY

Senior Backend & Cloud Engineer with 6+ years building scalable, cloud-native platforms on AWS and GCP. Deep expertise in Go, Python, and Kubernetes — focused on infrastructure reliability, cost optimization, and developer tooling. Proven track record leading platform features from POC to GA in multi-cloud, multi-team environments, consistently reducing operational costs and improving deployment reliability.

TECHNICAL SKILLS

Languages & DBs: Python, Go, Java, Bash, MySQL, PostgreSQL

AWS: VPC, EC2, RDS, S3, Lambda, SQS, API Gateway, IAM, Elastic Beanstalk, CloudFront

GCP: GKE, Compute Engine, Cloud DNS, Load Balancer, Filestore, Cloud Monitoring

Infra & Containers: Kubernetes, Docker, Helm, Crossplane, IaC, Terraform

CI/CD & Monitoring: GitLab, GitHub, Jenkins, Prometheus, Cloud Alerts

Practices: System Design, REST APIs, Security Hardening, Cost Optimization, Agile/Scrum

AI SKILLS

AI Tools & IDEs: Claude (claude.ai, Claude Code), GitHub Copilot, ChatGPT, Cursor

LLM APIs: Anthropic Claude API, OpenAI API — prompt engineering, structured outputs

AI-Assisted Dev: Code generation, test writing, documentation, refactoring using LLM tools

Agentic Workflows: Multi-step AI task automation, MCP (Model Context Protocol) integrations

Applied AI: RAG pipelines, AI-powered tooling, integrating LLMs into developer workflows

PROFESSIONAL EXPERIENCE

Senior Software Engineer | **HCL Software, USA** Feb 2024 – Present

Go, Python, PostgreSQL, Docker, Kubernetes, Helm, GKE, Filestore, Compute Engine, Crossplane, IAM

- Engineered a **hot sandbox provisioning system** that maintains a pool of pre-deployed sandboxes and dynamically assigns them to matching customer requests, reducing sandbox provisioning time by 75%.
- Designed and implemented auto-suspend system via request-listener and suspend-resume APIs, reducing idle resource costs across sandbox environments by ~25%
- Led VM-based deployment framework from POC to GA (v1.0.0), enabling hybrid Kubernetes/VM product deployments for multiple product teams without service disruption
- Improved platform reliability through secure sandbox deletion, VPA integration, and optimized cluster sizing, reducing over-provisioned resource spend by ~20%
- Enhanced platform security by implementing workload identity and streamlining upgrades for Keycloak, ACS PostgreSQL, Cert-manager, and Prometheus
- Authored reusable Terraform modules for GCP infrastructure provisioning, standardizing environment setup across sandbox and hybrid deployment workflows
- Designed and built a custom agent plugin framework, reducing operational burden and enabling product teams to self-serve deployments with clear separation of concerns
- Extended Crossplane GCP provider labels and optimized admin cost-tracking reports to support per-user data retrieval, improving billing visibility
- Collaborated across DevOps, Content, and QA teams to resolve critical production bugs and expand automated test coverage

Software Engineer (Intern) | **HCL Software, USA** Jul 2023 – Feb 2024

- Extended sandbox infrastructure to support multi-region GCP deployments, removing regional capacity bottlenecks and improving platform scalability
- Strengthened application security by patching internal libraries and addressing CVE vulnerabilities before they reached production
- Implemented resilience and fault-tolerance improvements to meet product SLA requirements under high-load scenarios

Senior Software Developer | **HCL Technologies, India** Aug 2021 – Aug 2022

Python, Boto3, Flask, AWS, Pytest, Redis, MySQL, Celery

- Replaced legacy nosetests with pytest across the full codebase, unblocking CI pipelines and standardizing test tooling, cutting test setup time
- Eliminated dummy-data contamination in unit tests by architecting an SQLite-backed test strategy, improving test reliability and developer confidence
- Reduced metrics processing latency and resolved IPv6 location-fetching limitations by migrating from MySQL GSI to Maxmind mmdb
- Hardened the metrics system against AWS regional outages using Redis-backed deduplication, improving fault tolerance across distributed components
- Deployed ALB with single-instance configuration for Trial cloud provisioning, reducing infrastructure cost for low-tier customers
- Mentored team members and new hires on provisioning workflows and platform tooling, accelerating onboarding ramp-up time

Software Developer | **Kony Inc, India** Oct 2019 – Aug 2021

Python, Flask, Boto3, Celery, Alembic, AWS, IaC, Jenkins, Java, MySQL, AngularJS

- Led Python 2 → Python 3 / Boto3 migration of the full Provision codebase, a cross-team initiative recognized with a Team Award
- Automated AWS subaccount provisioning within the Organization account using assumed roles, eliminating key-based authentication and improving security posture
- Built Jenkins CI pipelines on EC2 for Build Verification Tests (Java), eliminating manual QA runs and reducing pre-release validation time by ~30%
- Automated microservices deployments for existing products using API Gateway, Lambda, and DynamoDB, reducing release time and manual handoffs
- Delivered end-to-end features, enhancements, and defect fixes for the Cloud Provision component using Python and AngularJS

Associate Engineer | **Kony Inc, India** Oct 2018 – Oct 2019

- Enhanced the Provision UI used by the Operations Team for managing cloud environments
- Designed workflows for updating AMIs and Node.js server solution stacks, enabling faster patching cycles
- Took ownership of Build Verification Tests and developed a new automated test suite for latest AMI validation

EDUCATION

M.S. in Information Technology Aug 2022 – Dec 2023

University of Cincinnati

B.Tech in Computer Science Aug 2015 – Apr 2019

Keshav Memorial Institute of Technology (JNTUH)

CERTIFICATIONS & AWARDS

- Oracle Certified Java 8 Associate
- Extramile Award, HCL Software – Sep 2025: VM support to enable Hybrid deployments
- Standout Performer, HCL Software – Aug 2024: Recognized for multi-geo platform delivery impact
- Team Award, Kony – Jun 2020: Led successful Provision codebase migration to Python 3 / Boto3
- Extra Mile Award, Kony – Dec 2019: Key role in BVT code ownership transition from Accounts to Provision
- Spot Award, Kony – Jan 2020: Delivered Temenos Cloud go-live on schedule